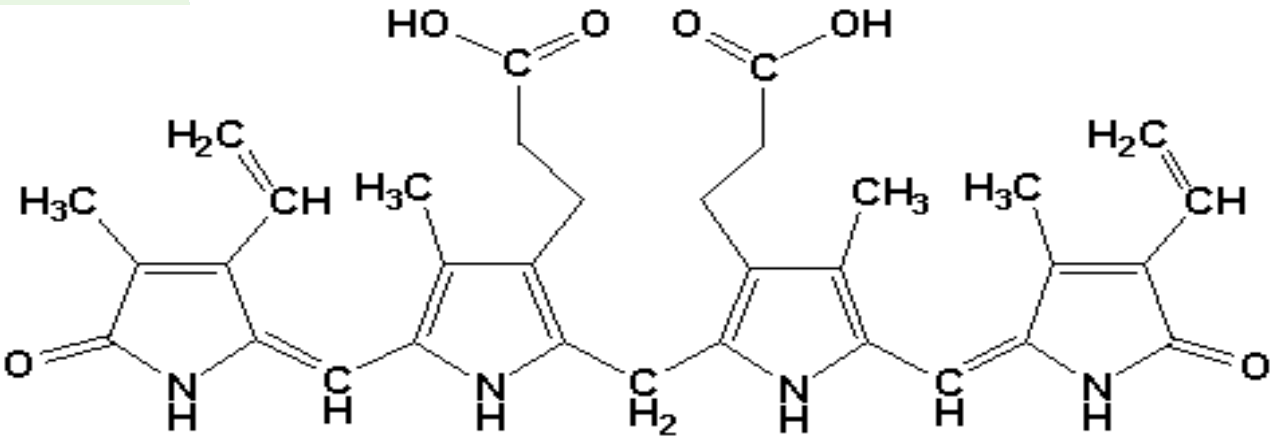
Interpretation of Bilirubin

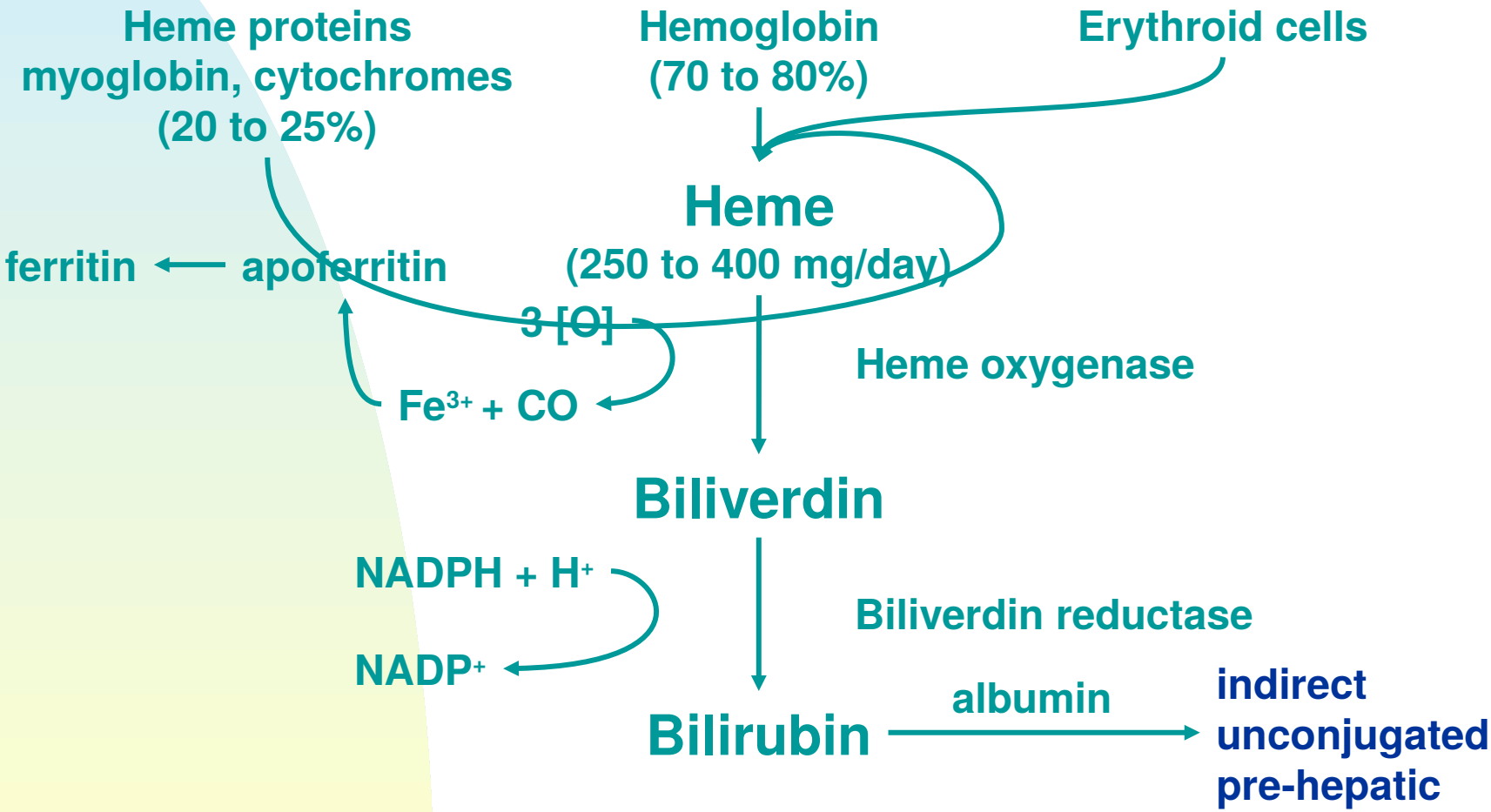
What is Bilirubin?

- Breakdown product of heme metabolism
- Bile pigment and also excreted in urine.
- Lipophilic in nature (insoluble in aqueous medium).
- It consists of an open chain of four pyrrole-like rings (tetrapyrrole)

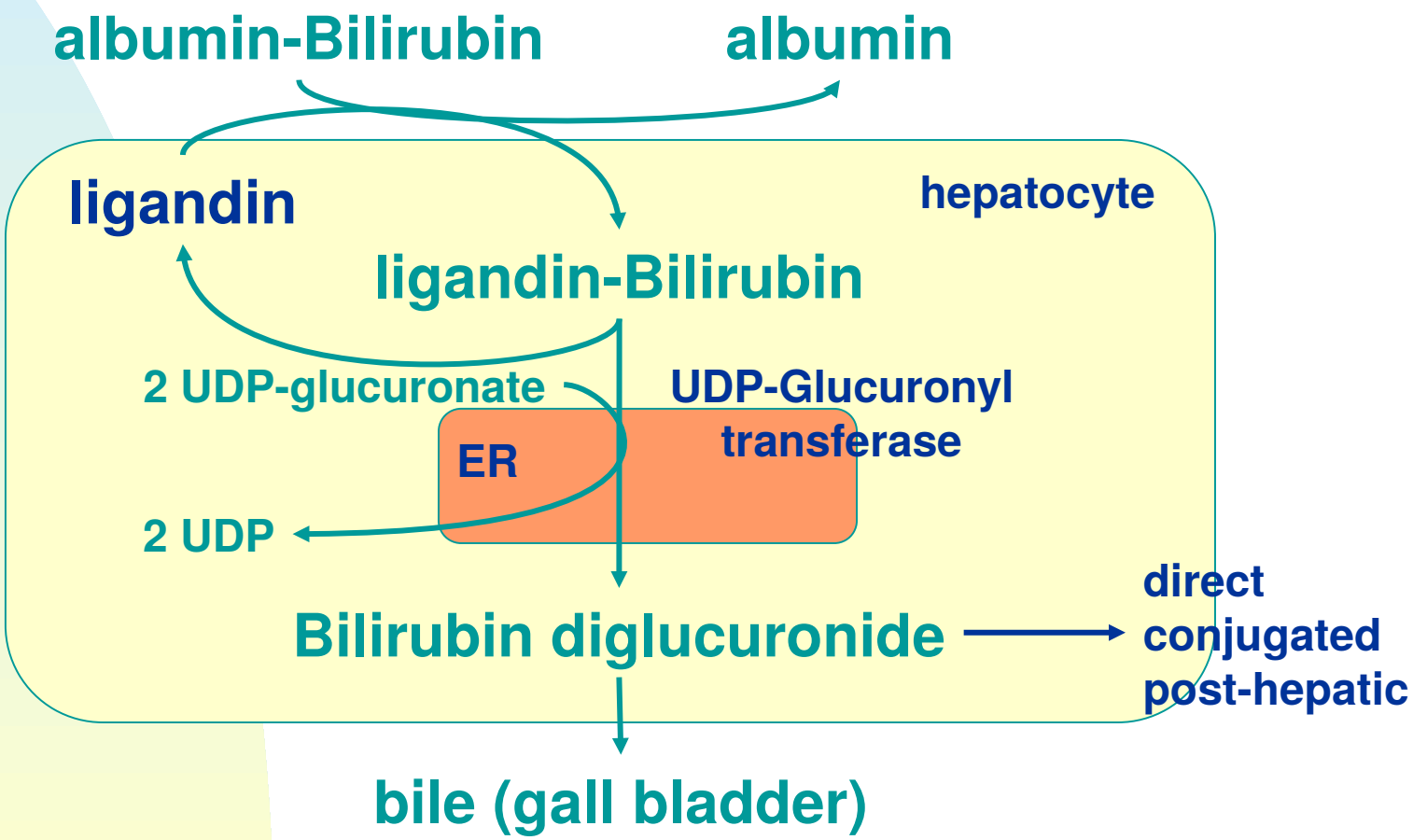


Bilirubin

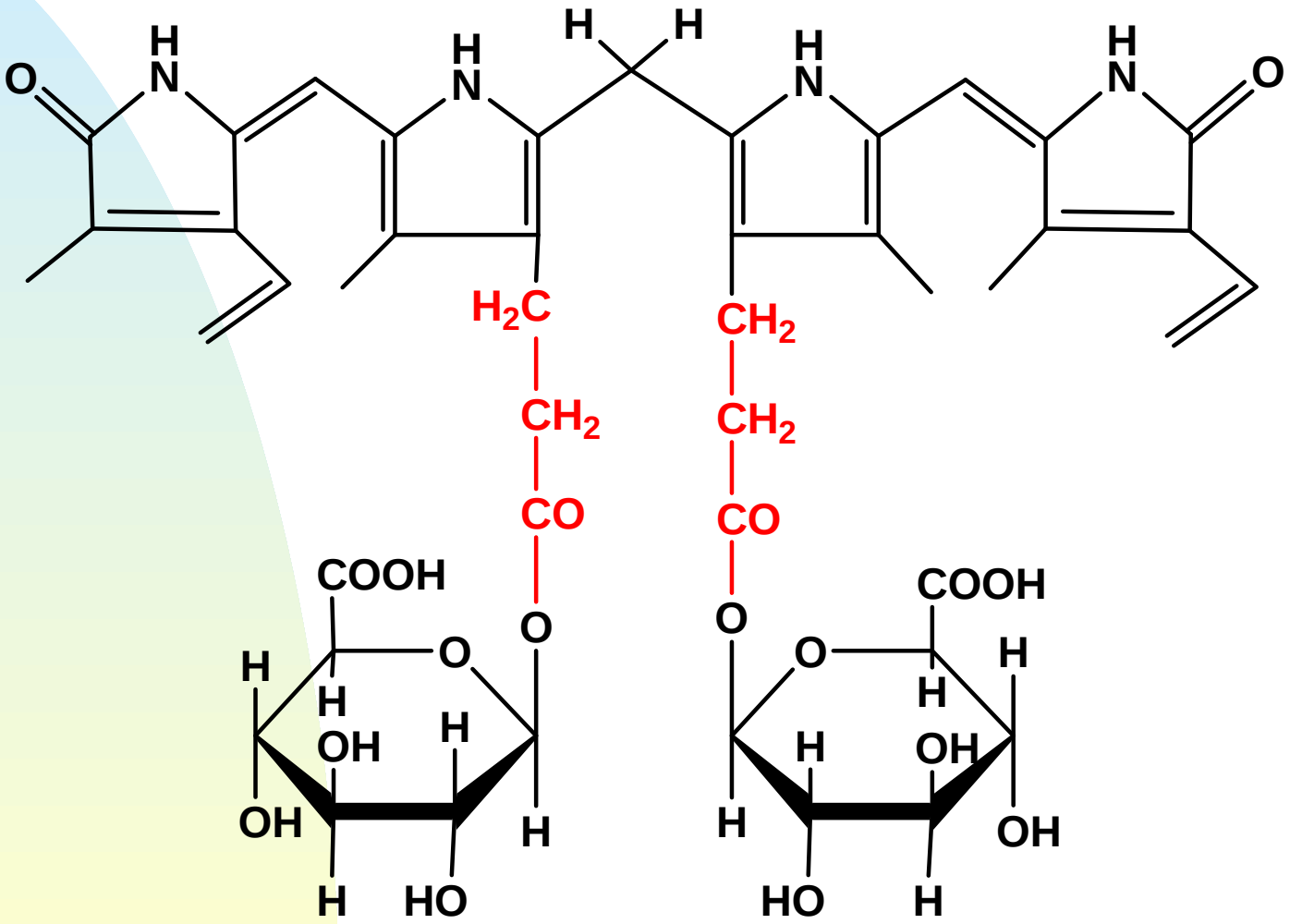
Bilirubin Production



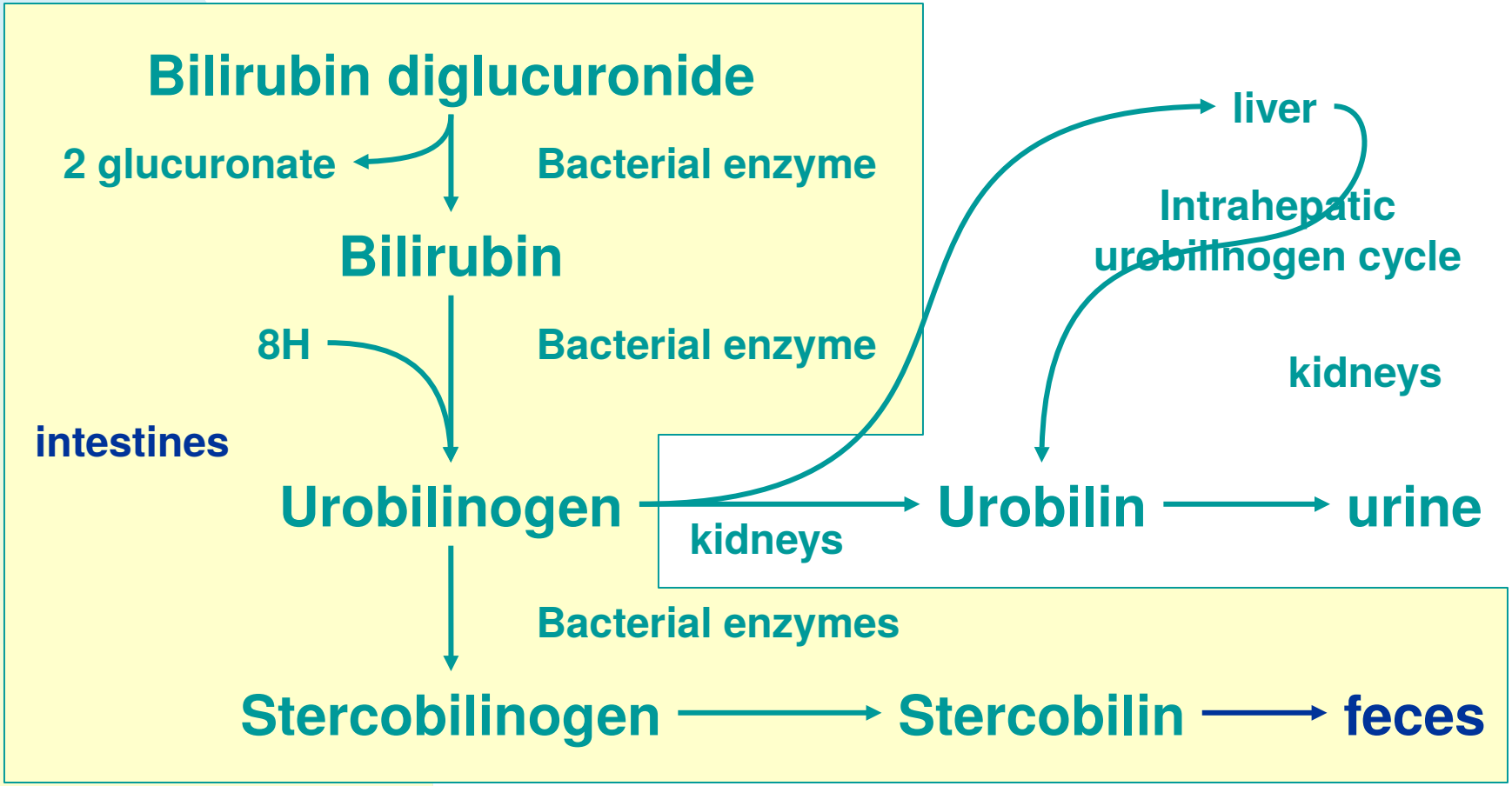
Bilirubin Processing



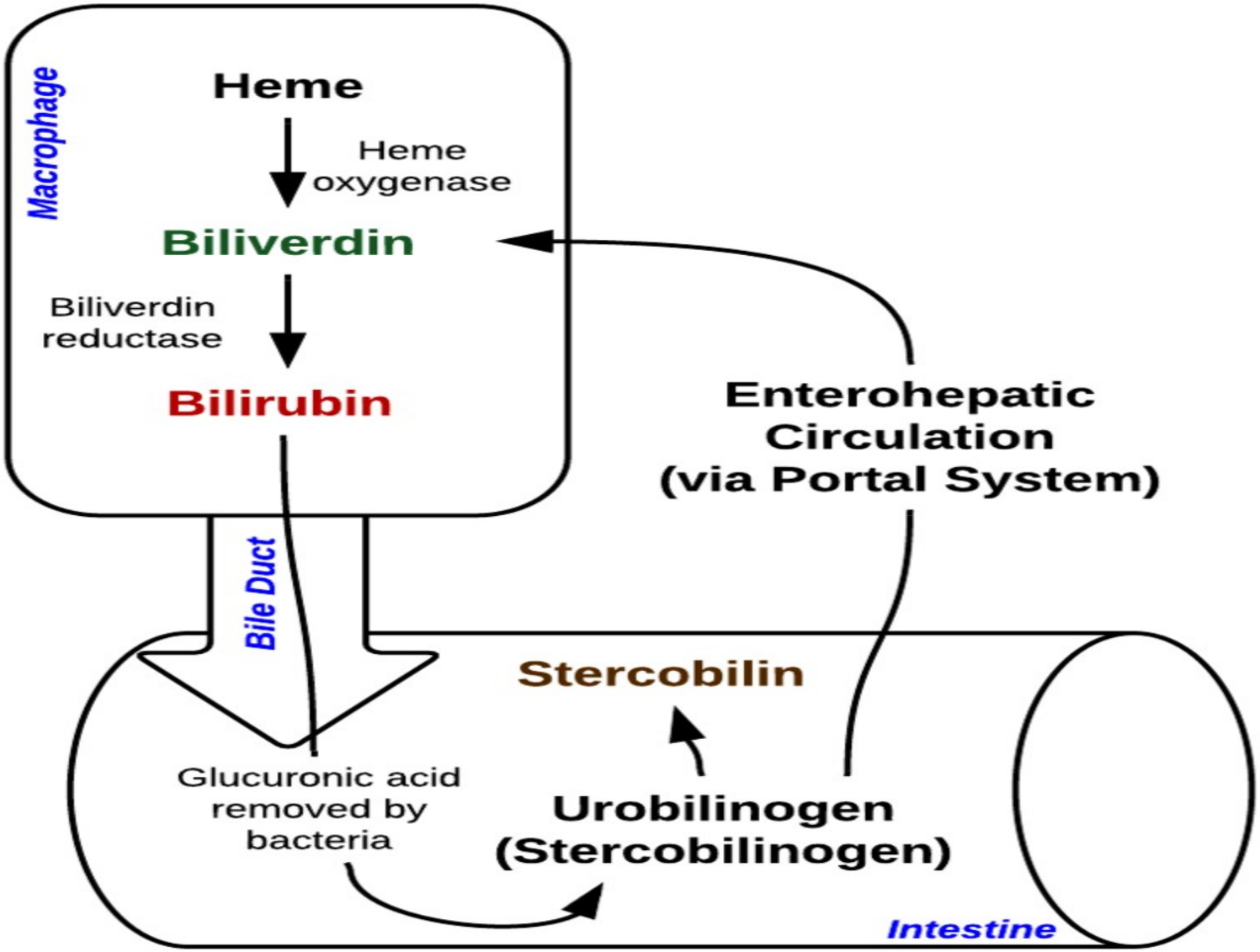
structure of bilirubin diglucuronide



Bilirubin Excretion



- Because bilirubin is highly insoluble in water, it must be converted into a soluble conjugate before elimination from the body.
- In the liver, uridine diphosphate (UDP)-glucuronyl transferase converts bilirubin to a mixture of monoglucuronides and diglucuronides, referred to as conjugated bilirubin, which is then secreted into the bile by an ATP-dependent transporter.
- This process is highly efficient under normal conditions, so plasma unconjugated bilirubin concentrations remain low.


Macrophage
Heme
Heme oxygenase
Biliverdin
Biliverdin reductase
Bilirubin
Enterohepatic Circulation (via Portal System)
Bile Duct
Stercobilin
Glucuronic acid removed by bacteria
Urobilinogen (Stercobilinogen)
Intestine

Bilirubin Lab values

Bilirubin form	Normal value
Total (elderly, adult, child)	0.1 to 1.0 mg/dL
(newborn)	1.0 to 12.0 mg/dL
Critical value (adult)	>12 mg/dL
(newborn)	>15 mg/dL
Pre-hepatic, unconjugated, indirect	0.0 to 0.8 mg/dL
Post-hepatic, conjugated, direct	0.0 to 0.25 mg/dL
Fecal urobilinogen	40 to 280 mg/day
Urine	0.0 to 0.02 mg/dL

Conjugated bilirubin - water soluble - direct reaction with dyes

Unconjugated bilirubin - water insoluble - alcohol is needed for dye (indirect) reaction

Observe the color changes associated with heme degradation by watching the progress of a bruise (dark red to green to yellow).

Difference of two bilirubins

	free bilirubin	conjugated bilirubin
Binding with Glucuronic acid	no	yes
Reacting with the diazo reagent	Slow and indirect	Rapid and direct
solubility in water	small	large
Discharged via kidney	no	yes
Pass through the membrane of cell	yes	no

Hyperbilirubinemia

- Interferences at any one of the points of bilirubin processing described above can lead to a condition known as HYPERBILIRUBINEMIA.

- As the name implies this disease is characterized by abnormally elevated levels of bilirubin in the blood.

SYMPTOMS

- Yellowing of the skin, scleras (white of the eye), and mucous membranes (jaundice)
- Detectable when total plasma bilirubin levels exceed 2mg/100mL



Causes

1. Increased bilirubin production
2. Reduced bilirubin uptake by hepatic cells
3. Disrupted intracellular conjugation
4. Disrupted secretion of bilirubin into bile canaliculi
5. Intra/extra-hepatic bile duct obstruction

Lead to increases in free (unconj.) bilirubin

Result in rise in conj. bilirubin levels

Jaundice

Definition : Accumulation of bilirubin or its conjugates in body tissues produces jaundice (ie, icterus), which is characterized by high plasma bilirubin levels and deposition of yellow bilirubin pigments in skin, sclera, mucous membranes, and other less visible tissues.

In laboratory - the Van den Bergh reaction was used for a qualitative estimate of bilirubin.

'Serum bilirubin reacted with diazotized sulfanilic acid (alcholic)'.

Classification of Jaundice

1. **Hemolytic Jaundice:** increased hemolysis of erythrocytes. Other causes may include sickle cell anemia, malaria, incompatible blood transfusion.

- Over production of bilirubin.
- Elevated levels of serum unconjugated bilirubin.
- More bilirubin is excreted into the bile – increased formation of urobilinogen & stercobilinogen.
- Dark brown color of feces (high stercobilinogen).

2. **Hepatic (hepatocellular) Jaundice:** damage to parenchymal cells caused dysfunction of liver.

- Major causes includes liver cirrhosis, viral hepatitis, hepatic damage associated with poisons or toxins.
- It adversely affects bilirubin uptake and its conjugation by liver cells.
- Elevated levels of conjugated and unconjugated bilirubin.
- Increase ALT & AST (released in circulation due to damage hepatocytes).

3. **Obstructive Jaundice:** due to the obstruction in the bile duct. It prevents the passage of bile into the intestine.

- It may be caused by gall stones, tumor, etc.
- In this case, the conjugated bilirubin from the liver enters the circulation.
- Elevated levels of serum conjugated bilirubin.
- Impairment in fat digestion & absorption (bile absence).
- Urine color is dark – elevated excretion of bilirubin.
- Clay color feces – absence of stercobilinogen.

Laboratory results in patients with jaundice

	normal	Hemolytic jaundice	Hepatocellular jaundice	Obstructive jaundice
Serum bilirubin				
total	< 1mg/dl	> 1mg/dl	> 1mg/dl	> 1mg/dl
direct	0~ 0.8mg/dl		↑	↑↑
indirect	< 1	↑↑	↑	
Urine bile pigments				
urobilirubin	–	–	+ +	+ +
urobilinogen	A few	↑	uncertainty	↓
urobilin	A few	↑	uncertainty	↓
Color of feces	normal	dark	Simple or normal	Clay color

Neonatal jaundice: Hyperbilirubinaemia accumulation of bilirubin (unconjugated) in certain brain regions.

Resulted in hyperbilirubinemic toxic encephalopathy or ***kernicterus***, that causes mental retardation, manifesting as various neurological deficits, seizures, abnormal reflexes and eye movements.

DISRUPTED INTRACELLULAR CONJUGATION (unconj. Hyperbilirubinemia)

- Crigler-Najjar Syndrome, Type I (CN-I) : congenital non-hemolytic jaundice. Recessive allele; mutation-induced loss of conjugating ability in the critical hepatic enzyme ***glucuronosyltransferase***
- CN-II : rare hereditary disorder (due to severe defect in bilirubin conjugation).

Greatly reduced but detectable glucuronosyltransferase activity due to mutation (predominantly recessive); enzymatic activity can be induced by drugs

- Gilbert's Syndrome

glucuronosyl transferase activity reduced to 10-30% of normal; also accompanied by defective bilirubin uptake mechanism

DISRUPTED SECRETION OF BILIRUBIN INTO BILE CANALICULI (conj. Hyperbilirubinemia)

- **Dubin–Johnson Syndrome**

  mild conjugated hyperbilirubinemia, but can increase with concurrent illness, pregnancy, and use of oral contraceptives; otherwise asymptomatic Inability of hepatocytes to secrete CB after it has formed due to mutation in the MRP2 gene (autosomal recessive trait)

- **Rotor Syndrome**

- **Autosomal recessive condition characterized by increased total bilirubin levels due to a rise in CB caused by a defect in transport of bilirubin into bile**

Thank you